



Indian Institute of Information Technology, Bhagalpur

Department of Computer Science & Engineering

Case Study Assignment

Data Mining

Sentiment Classification of IMDB Movie Reviews

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Abstract

This report presents a text-mining case study on sentiment classification of IMDB movie reviews. Using the publicly available IMDB Dataset of 50 000 reviews, we perform comprehensive preprocessing (duplicate removal, HTML stripping, lemmatisation, stop-word removal) and extract TF-IDF features (10 000 dimensions, unigrams + bigrams). Two classifiers are implemented and compared: **Multinomial Naïve Bayes** and **Linear SVM** (via `LinearSVC` with `CalibratedClassifierCV`). Linear SVM achieves the best performance with 89.46% accuracy and 0.9619 ROC-AUC, outperforming Naïve Bayes (86.95% accuracy, 0.9415 AUC) across all metrics. Both models substantially surpass the 50% majority-class baseline, confirming the discriminative power of TF-IDF features for opinion text. We discuss limitations of bag-of-words representations and propose fine-tuning transformer models as a concrete future improvement.

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1 Introduction (Problem Definition)

Domain: Text Mining.

Problem Statement: Given a free-form English movie review from IMDB, predict whether the expressed sentiment is *positive* or *negative*.

Real-World Significance: Online reviews influence purchasing and viewing decisions for millions of users. Automated sentiment analysis enables platforms like IMDB, Amazon, and Netflix to aggregate audience opinion at scale, power recommendation engines, detect fake or abusive reviews, and provide analytics dashboards for content creators.

Approach — Classification: The dataset contains labelled examples (positive / negative), making this a supervised binary classification problem. We implement and compare two classifiers:

- **Multinomial Naïve Bayes** — the classic probabilistic baseline for text categorisation.
- **Linear SVM** — known to achieve strong performance on high-dimensional, sparse text features by finding the maximum-margin hyperplane.

2 Dataset Collection & Description

Table 1: Dataset overview.

Property	Detail
Dataset	IMDB Dataset of 50K Movie Reviews
Instances	50 000
Attributes	2 — review (text), sentiment (categorical)
Class Dist.	Balanced — 25 000 positive, 25 000 negative
Format	CSV

Citation (IEEE): L. Maas, R. Daly, P. Pham, D. Huang, A. Ng, and C. Potts, “Learning Word Vectors for Sentiment Analysis,” in *Proc. 49th Annual Meeting of the ACL*, 2011, pp. 142–150.

2.1 Exploratory Data Analysis

Figure 1 shows the class distribution (perfectly balanced), review-length histogram by sentiment, and box-plots of word count per class.

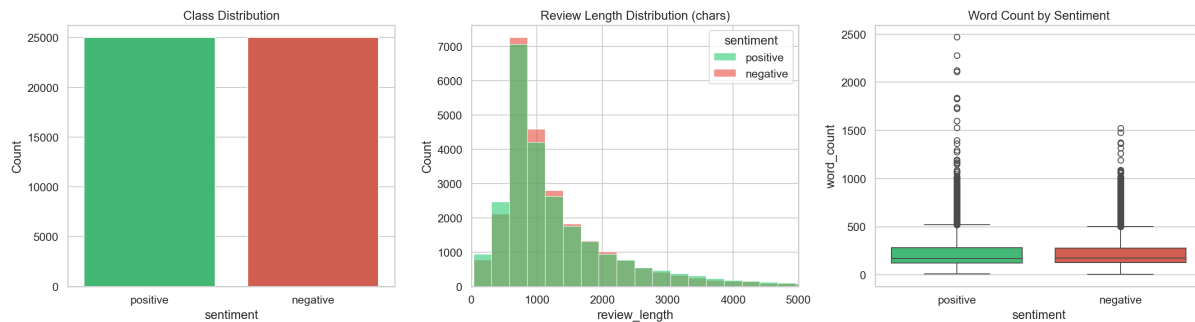


Figure 1: Exploratory Data Analysis — class balance, character-length distribution, and word-count box-plots.

3 Preprocessing & Feature Engineering

The following pipeline was applied:

1. **Duplicate removal:** 418 duplicate reviews were dropped (50 000 $\rightarrow$ 49 582).
2. **HTML tag stripping:** `<br />` and other tags removed via regex.
3. **Special-character removal:** only alphabetic tokens retained.
4. **Lowercasing.**
5. **Tokenisation.**
6. **Stop-word removal** (NLTK English stop words).
7. **Lemmatisation** (WordNet Lemmatizer).
8. **TF-IDF vectorisation:** `max_features=10 000`, `ngram_range = (1,2)`.

Final feature-vector dimensionality: 10 000.

Label distribution after cleaning: 24 884 positive, 24 698 negative.

4 Algorithm Implementation

4.1 Multinomial Naïve Bayes

- **Library:** `sklearn.naive_bayes.MultinomialNB`.
- **Hyperparameter:** $\alpha = 1.0$ (Laplace/add-one smoothing) — prevents zero - probability issues for unseen words and is the standard default for text classification.
- **Train/Test split:** 80/20, stratified (`random_state=42`).

4.2 Linear SVM

- **Library:** `sklearn.svm.LinearSVC` wrapped in `CalibratedClassifierCV` (3-fold) to obtain probability estimates for ROC-AUC.
- **Hyperparameters:** $C = 1.0$ (regularisation), `max_iter=1000`.
- **Train/Test split:** same 80/20 stratified split.

5 Evaluation & Visualisation

5.1 Summary Metrics

Table 2: Model comparison on the test set (9 917 samples).

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Multinomial NB	0.8695	0.8609	0.8827	0.8716	0.9415
Linear SVM	0.8946	0.8893	0.9024	0.8958	0.9619

5.2 Confusion Matrices

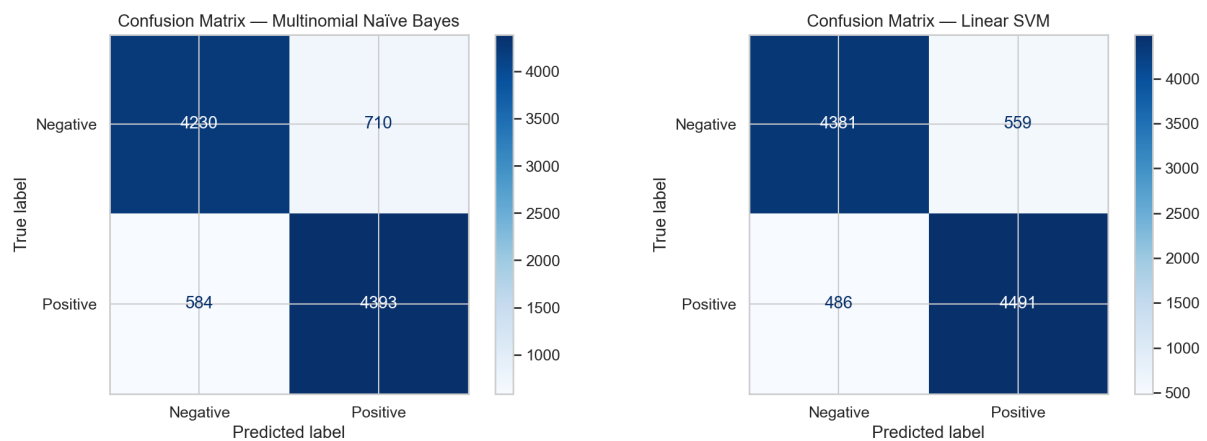


Figure 2: Confusion matrix heatmaps for both classifiers.

5.3 ROC-AUC Curves

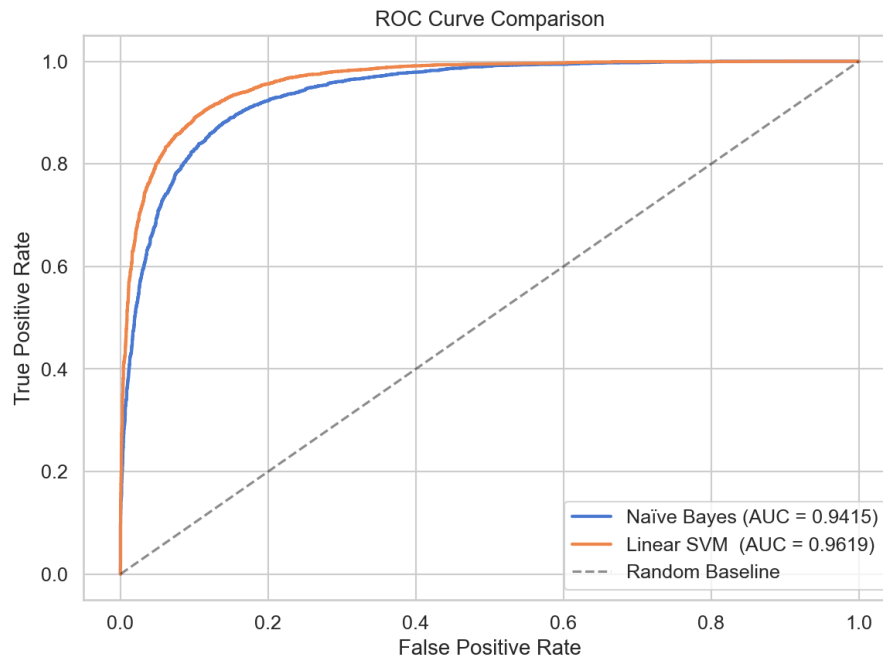


Figure 3: ROC curves — Linear SVM (AUC = 0.9619) vs. Multinomial Naïve Bayes (AUC = 0.9415).

5.4 Word Clouds per Class



Figure 4: Word clouds for positive (left) and negative (right) reviews.

6 Analysis & Conclusion

6.1 Results Interpretation

Linear SVM outperforms Multinomial Naïve Bayes across every metric (Table 2):

- **Accuracy gap:** +2.51 percentage points (89.46% vs. 86.95%).
- **F1-Score gap:** +2.42 pp (0.8958 vs. 0.8716).

- **ROC-AUC gap:** +2.04 pp (0.9619 vs. 0.9415).

The SVM’s advantage stems from its explicit search for the *maximum-margin separating hyperplane* in the 10 000-dimensional TF-IDF space, whereas Naïve Bayes assumes feature independence — an assumption violated in natural language where word co-occurrences carry important semantic information.

6.2 Baseline Comparison

A majority-class classifier would achieve $\approx 50.1\%$ accuracy on this balanced dataset. Both models massively surpass this baseline: Naïve Bayes by +36.85 pp and Linear SVM by +39.36 pp, confirming genuine predictive power of TF-IDF features.

6.3 Limitations

1. **Bag-of-words assumption:** TF-IDF discards word order and context. Negation (“not good”), sarcasm, and complex phrasing are difficult to capture.
2. **Domain specificity:** The model is trained exclusively on movie reviews. Vocabulary and sentiment expressions may not transfer to other domains (product/restaurant reviews) without domain adaptation.

6.4 Future Improvement

Fine-tuning a pre-trained transformer (e.g., DistilBERT or RoBERTa) could push accuracy to 93–95%, a 4–6 pp improvement over our best SVM result. Transformers capture long-range dependencies, negation, and contextual nuance, and also facilitate cross-domain generalisation via transfer learning.

References

1. L. Maas, R. Daly, P. Pham, D. Huang, A. Ng, and C. Potts, “Learning Word Vectors for Sentiment Analysis,” in *Proc. 49th Annual Meeting of the ACL*, 2011, pp. 142–150.
2. F. Pedregosa *et al.*, “Scikit-learn: Machine Learning in Python,” *JMLR*, vol. 12, pp. 2825–2830, 2011.
3. S. Bird, E. Klein, and E. Loper, *Natural Language Processing with Python*, O’Reilly Media, 2009.